

## Distance and midpoint



- 1      a   12.8 units                      b   6.2 units                      c   5.9 units                      d   14.5 units
- 2      a    $(-0.5, 1.5, 2)$                       b    $(3.5, -2, -2)$                       c    $(-5, 7, 0.5)$                       d    $(14, 8.5, 10)$
- 3      a      i   6.48 units                      ii    $(0.5, 0.5, 1)$                       b      i   12.69 units                      ii    $(1, 3.5, -2)$
- c      i   10.10 units                      ii    $(4.5, -6, -0.5)$
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## Applications

- 4      a   4 units                      b    $96 \text{ units}^2$                       c    $64 \text{ units}^3$
- 5    $729 \text{ units}^3$
- 6    $174.9 \text{ units}^2$